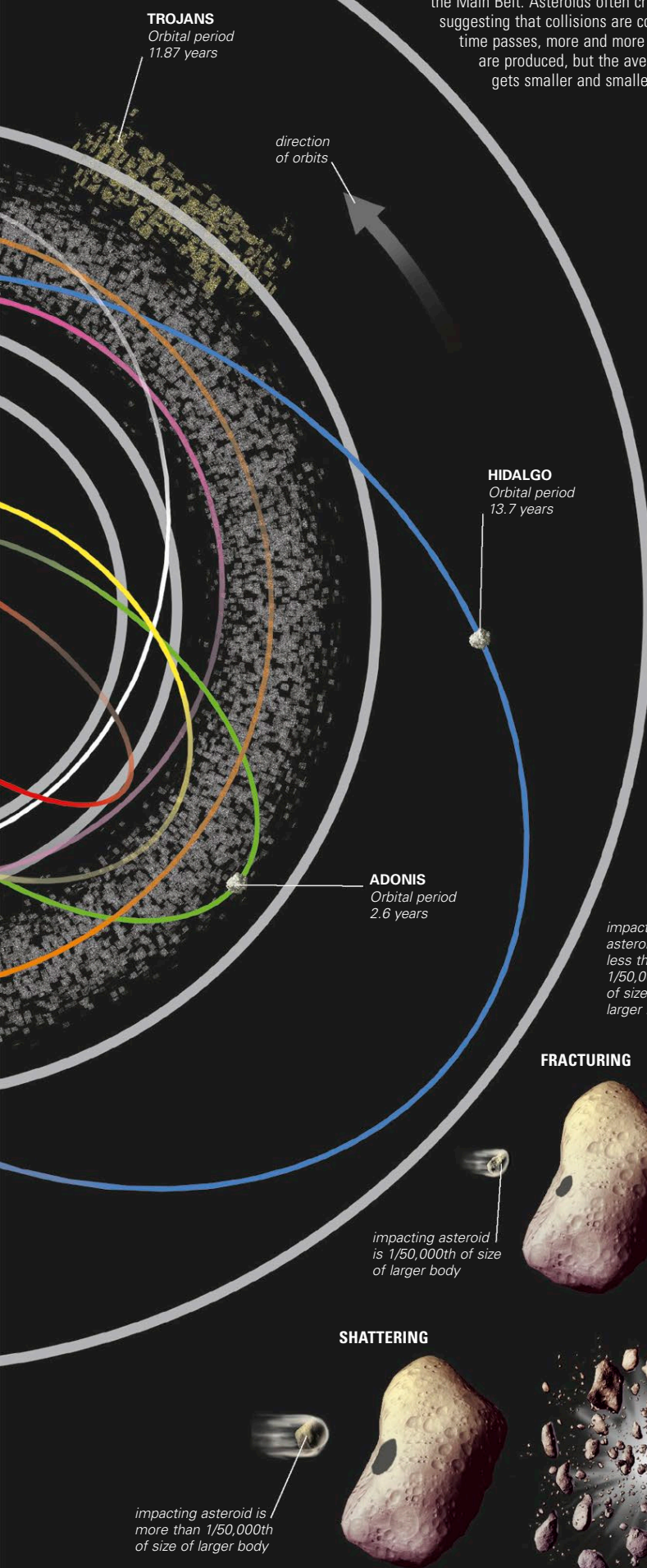
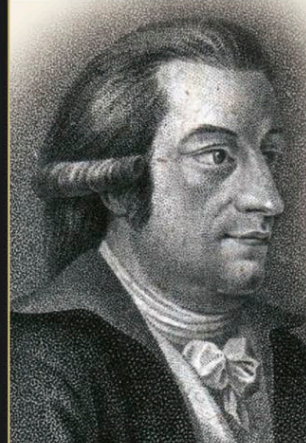


ASTEROID ORBITS

Asteroids tend to stay close to the plane of the solar system, and they orbit in the same direction as the planets. A few individual asteroid orbits are shown here, together with the Main Belt. Asteroids often cross paths, suggesting that collisions are common. As time passes, more and more asteroids are produced, but the average size gets smaller and smaller.



FRANZ XAVER VON ZACH

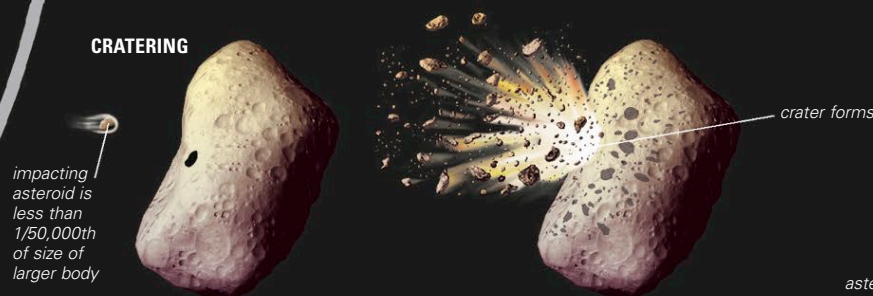


Franz Xaver von Zach (1754–1832) was a Hungarian baron and the director of the Seeberg Observatory in Germany. He became convinced that there was a missing planet orbiting between Mars and Jupiter. In September 1800, he organized a group of 24 astronomers to help him look. Popularly known as the Celestial Police, they divided the celestial zodiac into 24 parts and started searching but were beaten by a nose by Giuseppe Piazzi's accidental discovery of the asteroid Ceres in 1801. The Police were surprised by how small Ceres was, then were surprised again when more and more asteroids were found in similar orbits.

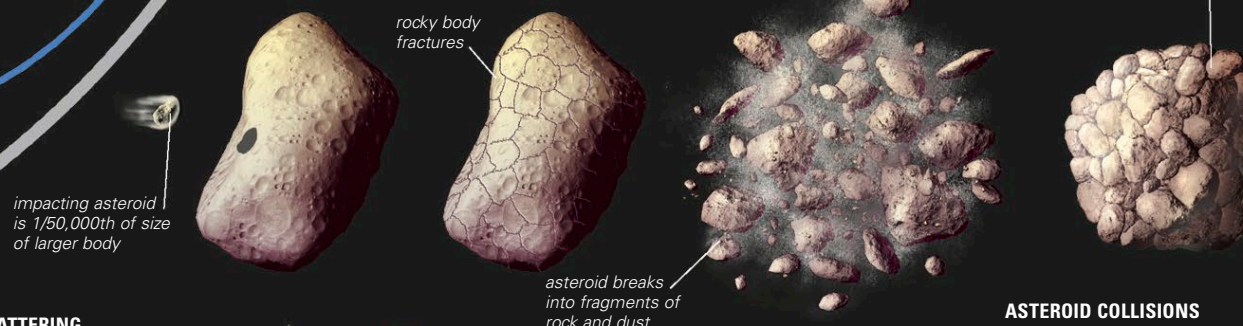
COLLISIONS

The effect of a collision between asteroids depends on the sizes of the bodies involved. If a very small asteroid hits a larger one, it will produce a crater on the surface. This crater will be about 10 times the size of the incoming body. Because asteroids are much smaller than planets, the material blasted out of the crater will escape and move off into an independent orbit around the Sun. This orbit will, however, be very similar to that of the impacted asteroid, and there is a good chance that the ejected material will hit the cratered asteroid again. A bigger impactor can break up the asteroid that it hits. But so much energy is used to do this that the resulting fragments cannot escape from the gravitational field, and they will all fall back to form an irregular ball of rubble. Subsequent minor impacts will break up the surface, covering the asteroid in a rocky, dusty layer. A casual observer would not realize that the underlying asteroid is actually in pieces. A large impactor will not only shatter the asteroid, but the fragments will also escape. These will form a family of asteroids that eventually spreads out around the orbit of the original body.

CRATERING



FRACTURING



SHATTERING



ASTEROID COLLISIONS

There are far more small asteroids than large ones. For every asteroid more than 6 miles (10 km) along its longest axis, there are 1,000 longer than 0.6 miles (1 km) and a million longer than 0.06 miles (0.1 km). So cratering is much more common than fracturing, which in turn is much more common than shattering. Asteroids that are shattered are likely to be already fractured.